

Autonomous Vehicles, Trolleys, and the Ethics of Machine Learning

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In this lecture, we take up the issue of whether and how designers of autonomous vehicles should approach programming their vehicles to respond in accident scenarios. Trolley Optimists believe that consideration of so-called trolley cases can and should be used to directly inform how autonomous vehicles should behave in such scenarios. The idea is that we can consider what we should do when, say, a runaway trolley is going to hit and kill five people but can be diverted to hit and kill only one and then program an autonomous vehicle to behave similarly. Trolley Pessimists believe it is a mistake to use trolley cases in this fashion. We outline some different versions of each of these positions, articulate some challenges to views on both sides, and then explain why careful consideration of the details of AI and machine learning justify a version of trolley pessimism.

Lecture Transcript

Basl 0:00 Hi, I'm John Basl. I'm an Associate Professor of Philosophy at Northeastern University and a member of the Northeastern Ethics Institute.

Behrends 0:09 And I'm Jeff Behrends. I'm Associate Senior Lecturer in the Department of Philosophy at Harvard University and director of ethics and technology initiatives at the Edmond J. Safra Center for Ethics.

Basl 0:20 In this video, we're going to talk about the ethics of autonomous vehicles and the ethics in particular of trying to draw on so-called trolley cases or trolley problems, to resolve a specific set of problems in the building or designing of autonomous vehicles. And in doing so, and in diagnosing some issues there, we're going to hopefully draw some broader lessons for the ethics of machine learning. I'm going to start with a thought experiment. I'd like to ask you to imagine Annabel. Annabel is a member of an engineering team that's responsible for programming the driving behavior of her company's autonomous vehicle model. She knows that the AV will at some point be in the kind of scenario now frequently discussed in ethics classrooms in the popular media. No matter how the car behaves, a collision is unavoidable risking harm to its occupants or others. Annabel is not interested in what is best for her company's bottom line. She intends to do what she is obligated to do, all things considered, that is taking all the reasons she has moral and otherwise into account, she's going to do what she has most reason to do. She remembers studying trolley cases in Intro to Philosophy. And we're going to talk in detail about some of those cases later, just if you're not familiar with them. And she's read some popular media articles making the comparison between so-called trolley cases where you're deciding how a runaway trolley should behave, and accidents that an autonomous vehicle might be in or accident scenarios that an autonomous vehicle

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might be in. And the question Annabel is thinking about is could attending to those trolley cases help her to understand what she and her colleagues ought to do. Now, what we want to do is try to clarify how theorizing about so-called trolley cases or those trolley problems does and does not relate to AV design. And then sort of learn some lessons from that about thinking about the ethics of machine learning or autonomous vehicle design more broadly. Now, before we proceed, a quick note about the central question of this paper, one might think that it's a mistake to focus on decision context like Annabel's, we should instead attend to a policy level question concerning the ethics of autonomous vehicles. Now, there are important policy questions concerning autonomous vehicles, but it would be a mistake to overlook Annabel's decision context entirely. This is because at least currently, policy does not constrain the options of car manufacturers in a significant way. And it's unlikely to do so anytime soon. And the choices made by engineers like Annabel are themselves morally significant. So even if they were constrained by some policy, they still have moral decisions to make. Now perhaps there will be a time at which the central question we're trying to address no longer matters because Annabel just won't have any Annabel or engineers like her just won't have any options to choose among, there'll be tight regulations telling them exactly how they should program autonomous vehicles to behave. But that time appears to be far off. And in the meantime, we should try to give whatever advice we can to engineers or engineering firms designing autonomous vehicles. Okay, now we can turn to the meat of the video or talk. Very roughly, we want to undermine or raise questions for a view that we call trolley optimism. That's the view on which thinking about trolley cases or trolley problems has immediate implications for thinking about how we should design AVs to behave. Basically, we think about the trolley cases, and they provide immediate information. They tell us what to do when we're programming autonomous vehicles for how they should behave in accident scenarios. We'll start by outlining more clearly what trolley optimism is. We will give a much careful, much more careful picture of that. And then we're going to distinguish between two different forms of trolley optimism. One is called democratic trolley optimism and the other we're going to call standard trolley optimism and we'll get into those details. We're then going to raise some problems that are specific to the democratic variant of trolley optimism. And then we'll turn to the more prevalent, at least within philosophy and philosophical discussions, standard form of trolley optimism. We're going to consider one objection the literature raised against this form of optimism, and try to show that it might not undermine the trolley optimists successfully. Then we're going to turn to our objection, which we call the technological challenge to trolley optimism. We'll explain that, we'll explain that challenge to trolley optimism. We will take up some objections and replies that'll help clarify our view. And we'll use that to draw out some implications for theorizing about the ethics of AV design and the ethics of machine learning more generally. I'm going to turn it over to Jeff and he's going to more carefully characterize trolley optimism and some of the objections to it.

Behrends 4:49 As John said, I'm going to spend some time explaining how we understand trolley optimism. So roughly that idea is that autonomous vehicles will sometimes be in scenarios that look a lot like scenarios that have been proposed and variations of the trolley case thought experiment. And further, the idea is that we can use the trolley cases to figure out how the autonomous vehicle should behave in their accident scenarios. More precisely, we take trolley optimism to be subscription to the argument that's in front of you. We're the kinds of philosophers who like looking at arguments and premises and conclusion forms. So that's what we've done here. And importantly, trolley optimism is not just an endorsement of this conclusion, but of the particular inference and reaching the conclusion. So let's walk through this argument and unpack it a little bit. Premise one says that some accident scenarios are structurally similar to trolley cases. What a structural similarity amounts to roughly the idea

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is that an accident scenario will be structurally similar to a trolley case, when there's the same number of possible routes for each kind of vehicle to travel, the same moral consequences follow for each route traveled, and the same causal process for effecting those consequences is present in both cases. Let's illustrate that quickly by thinking about two famous variations of the trolley case. In the standard version, the trolley is moving down the track and if it isn't stopped or moved, it will strike and kill five workers. There's a sidetrack and if we pull the switch, it'll be diverted on to that sidetrack that will strike and kill one worker. It's easy to imagine an autonomous vehicle scenario that will be structurally similar. The car's moving ahead, if it doesn't swerve, it'll strike and kill five pedestrians. If it does, it'll strike and kill one. But other variations of the trolley cases are a little hard to imagine. So let's, let's walk through just one more. In another famous version of the trolley case, a trolley is moving down the track. And again, if nothing is done, we'll strike and kill five people. But now the only way to stop that is to push a very large person in front of the trolley. If we do that, it will strike and kill the large person and then derail, so the five are spared. What would it look like for an accident scenario to be structurally similar to that trolley case? Here we have to be a little more imaginative. Maybe the situation is something like this, the autonomous vehicle's approaching an intersection. On the other side of the intersection, five pedestrians have unexpectedly leapt into the crosswalk. There's no time for the autonomous vehicle to decelerate and avoid the collision. But a second autonomous vehicle is approaching horizontally and approaching the same intersection. Maybe a shared network could speed that unmanned autonomous vehicle up so that it collides with the first autonomous vehicle striking and killing the one passenger in the first vehicle but avoiding collision with a five. Maybe that's what it would look like for that case to be structurally similar to the second variation of the trolley case. When we'll leave it to your creativity and imagination to think of structural similarity for other cases. Now on to premise two. Premise two simply says that, if that's right, if accident scenarios are structurally similar, similar to trolley cases, then engineers should work to program autonomous vehicles such that they behave in those accident scenarios in ways that conform to the appropriate verdict and the relevant trolley cases. In what follows we're going to say a little bit about what conforming to appropriate verdicts amounts to. And we'll see that depending on how we interpret that, we can uncover at least two versions of trolley optimism. According to standard trolley optimism, the appropriate verdicts to guide programming decisions are simply the correct verdicts and trolley cases that are relevantly similar to accident scenarios. So what an engineer like Annabel should do is to program her autonomous vehicle so that in accident scenarios the car's behavior conforms with the moral obligations that a person would have in directing a trolley in relevantly similar trolley cases. So again, imagine the standard version. If it's true, for instance, that you're obligated to pull the switch, diverting the trolley away from the five and toward the one, then Annabel is obligated to see to it that her autonomous vehicle would behave similarly in a structurally similar scenario. This version of trolley optimism is probably the one that's most familiar to you of the two that we'll talk about, because it's the one most frequently appealed to either explicitly or implicitly, in both academic and media discussions of AVs and the trolley problems.

Behrends 9:33 But it isn't the only version of trolley optimism. Democratic trolley optimism differs from its cousin by offering a different set of trolley case verdicts that are relevant for programming AVs. According to democratic trolley optimism, autonomous vehicles should be programmed so that their behavior and accident scenarios conforms to group preferences about how they should behave. To illustrate that idea more fully, let's consider a recent proposal that aims to make use of data from an online platform called Moral Machine, a project developed at MIT Media Lab. Moral Machine is designed for two purposes. Here's a quote from the site describing those purposes: quote, "this website aims to take the discussion further by providing a platform for

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one, building a crowdsourced picture of human opinion on how machines should make decisions when faced with moral dilemmas and two, crowdsourcing assembly and discussion of potential scenarios of moral consequence” end quote. It accomplishes the first of those goals by presenting users with a series of cases involving accident scenarios with two possible outcomes. Users are asked to judge quote, “what the self-driving car should do”, end quote. It accomplishes the second goal by allowing users to construct their own cases selecting from a range of possible variables offered by the site. In a recent paper, some researchers have argued that the data collected by Moral Machine can be used to answer moral questions about the behavior of autonomous vehicles. The authors propose the following methodology. We should: one, collect data via Moral Machine; two, use that data to create models of the preferences of individual users of all possible cases; three, from those individual models create a single model meant to capture the group preferences for all possible cases; and finally four, use that group model to determine vehicle driving behavior. In short, the proposal offers a particular voting aggregation method for using the individual preference models to determine what the group prefers. And then guiding vehicle behavior accordingly. We’ll raise two sets of objections to the democratic trolley optimism approach just described. First, there are reasons to doubt the moral permissibility of relying on the data that the Moral Machine platform collects. At the data collection stage, we worry about the variables that are present in the cases. The image you see now shows the character variables you can choose among when building a crash scenario, including features like gender, body shape, and profession. Some of these features like body shape, for example, are obviously morally irrelevant. In an accident scenario, it doesn't matter whether there's a large person or a small person on one side of the intersection, for determining which way the car should swerve. But because the democratic approach works by having preference models vote, so long as individual users have preferences about differences that are genuinely morally insignificant, it's possible that the approach will result in encoding and treating as decisive group biases that we ought not have. At the modeling stage, we worry about treating the clicks of users as actually indicative of their preferences. Users may be clicking simply to move through cases, or are recording the results of a vote in a class or are running a pseudo experiment on their children. Additionally, users are not told that the data might be used in the way that the researchers propose. And finally, it's impossible to register indifference in the outcomes of these cases. For all of these reasons, some clicking behavior clearly does not correspond to actual user preferences. Now, we don't want to overstate this last point, perhaps the problems we've discussed here could be solved by more careful data collection design. And we certainly don't want to charge the researchers involved with unwavering allegiance to this exact data collected in this exact way. But we do think that these concerns are meaningful given the present experimental design. More significantly, though, it isn't clear why choices like Annabel’s should reflect group preferences in the first place. Absent some kind of moral argumentation for that view, it doesn't seem that we have reason to think that what private actors like Annabel or her company ought to do is maximize group preference satisfaction. Even if the Moral Machine platform did accurately reflect group preferences, and we could eliminate the risk of bias, we should still be careful not to treat the recorded preferences as immovable obstacles around which we have to navigate. Alternatively, we might want to embark on a public messaging campaign to change people's preferences, especially if they don't align with what would be socially valuable with respect to AV design.

Behrends 14:12 Democratic trolley optimism is explicit about its reliance on a voting mechanism among the preferences. Why think that a democratic decision making procedure of that kind is appropriate in this case? The points I've discussed here are related. The question we confront when presented with preference data is: what should we do in light of it? We might treat the preferences as fixed or not. And if we do treat them as fixed, we

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might prefer a democratic decision procedure over them, or not. Of course, this isn't to say that the preferences revealed by Moral Machine or other sources are in no way morally significant. Moral Machine researchers often point out in the popular press that it's important to know what people think of the behavior of AVs because it's important to know how they will respond to them as consumers. That seems entirely reasonable. But the democratic approach takes the group's preference as settling the moral questions confronting Annabel, and we shouldn't sign on to that idea, absent some convincing argument for it. So much for democratic trolley optimism. We'll turn now to assessing a significant objection to the optimists' argument. The disanalogy objection. As the name suggests, this objection is that trolley optimism of any sort is mistaken, because accident scenarios and trolley cases are not analogous. And this is problematic because the trolley optimist apparently takes them to be. The objection is most forcefully raised in a recent paper by Nyholm and Smids. I'm going to explain how the reasoning works. And then argue that, in fact, the reasoning isn't very good before handing things over to John who will present our own objection to trolley optimism. Nyholm and Smids identify three putative points of disanalogy between accident scenarios and trolley cases involving: first the decision making context, second liability and three uncertainty. Regarding decision making, they say that in the trolley scenarios, decision making happens immediately in isolation and with narrowly circumscribed information. Whereas none of that is true in accident scenarios, or at least in designing autonomous vehicles to encounter them. With respect to liability, both criminal and moral responsibility appear relevant in the AV context. But these considerations are often stipulated away in trolley cases. And finally, in accident scenarios, we have a great deal of uncertainty about what the environment will be like and how the car will respond in it. At best, we could get probabilistic conclusions about what those things will be like. Whereas in trolley case thought experiments, these things are often stipulated with certainty. On reflection, we think that none of these points gives us sufficient reason to be skeptical over relying on trolley cases to address questions like Annabel's. Let's address each in turn. With respect to decision making, we think that Nyholm and Smids have erred by mischaracterizing what we're doing when we're using trolley cases. They've described circumstances particular to the character in the case. The character has to decide quickly, they don't have time to confer with other people, and they only have a certain amount of information. Whereas what actually matters is our deliberative point of view when thinking about the cases. There is nothing at all that prevents us from reasoning together about the correct answers to all the cases or thinking about them for a long time. Indeed, this is just what philosophers do with them. We can even imagine that we're using the cases in perspective planning so that we're taking our time to settle some practical choice that we might face later on. We do think that Nyholm and Smids are right that there are important differences between what's going on in Annabel's decision making context and what's going on when we think about trolley cases. But they haven't given good reason to support that idea. John will revisit that idea when we present our favorite objection to trolley optimism. Now let's turn to liability and uncertainty. We think that appeals to these two features can both be addressed in the same way. Namely, by emphasizing that trolley cases don't have to exclude variables of these kinds. There's nothing special about the thought experiments that would require that we rule them out, even though popular cases usually do. The structure of the thought experiment seems almost maximally flexible, allowing us to tinker with a seemingly unlimited number of variables. So if we think for example, that criminal liability might matter morally in some way, we simply build a trolley case in which the character will be held criminally liable if they pull the switch or don't. That concludes our criticisms of the disanalogy objection. Now I'll hand things back over to John to explain why trolley optimism is really mistaken.

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Basl 18:48 We refer to our objection to trolley optimism as the technological objection, because the objection is grounded or gets its force from paying careful attention to one of the key technologies that power autonomous vehicles - machine learning. By paying attention to the ways in which machine learning is different from what we might call the traditional paradigm of programming, we can start to see the ethical questions we face are different from those that the trolley optimist urges us to confront and resolve by direct appeal to trolley cases. To start, algorithms in general are a mapping of input to output. In traditional programming, those used until recently in, you know, almost all software applications, inputs are mapped to outputs sort of by hand or specific outputs are explicitly specified step by step from a certain kind of input. Someone is providing an explicit recipe of how to get from some input to some output. Now, an example of the traditional approach, we can talk about a simple addition program. Given any two inputs, we explicitly programmed the system to do the following: sum them and then return the total. We could illustrate the point with many many other simple programs but even very complicated programs are created and have been created in this way. In contrast, a machine learner is itself an algorithm. But its output is an algorithm or more technically a model that itself, then map's inputs to outputs. So instead of explicitly defining outputs, we provide a learner with a collection of data called a training set. And we let it do the work itself of creating the input-output mappings that we might care about. So I think a useful analogy for creating input-output mappings using machine learning is dog training. In dog training, we can't just get inside a dog's brain and pair up commands with behaviors step by step. Instead, what we do is we set things up, we create an environment that's going to get the dog to do things. And then we give the dog commands and we reward the dog when it does the things we want, when it follows the commands. Now, what we hope is happening when we train a dog to sit is that it's writing internally an algorithm that takes spoken words like sit stay, paw, rollover, and outputting behaviors sitting, staying, pawing, rolling over. But we don't get to specify those by hand. Instead, we do it by training, and we let the dog itself write an internal algorithm. Now at first glance, it might seem like machine learning is a terribly inefficient way to create a program. After all, I wish I could hire a neuroscientist to come in and just write all the steps from my spoken command inputs to behavioral outputs, instead of having to go through all the hard work of training my dog. But despite programmers being very clever, some problems are just immune to traditional programming. If you want reliable, useful input-output mappings, machine learning is often the best tool we have. And it's made it possible to solve a range of problems, we couldn't solve very well using the traditional approach. And part of that comes because we have lots of data that we can supply to get good input-output mappings. For example, consider spam filtering. Imagine yourself trying to write an input-output mapping about what signs to look for in emails, those are the inputs, that would allow you to reliably classify those or get someone else to classify those as spam. Now, there's some low hanging fruit. If you get a random text of letter, a random string of letters or random capitalization, you might say well count, that's your input email, then counter to spam. But spam can be extremely sophisticated and sneak by such filters. So with a machine learner, we don't have to think about that problem at all. We can just take a machine learner, give it a huge set of spam emails and non spam emails that are marked and let it come up with an algorithm that it uses to classify them as spam or not. And we can even improve that algorithm over time. Every time you provide a market email as spam, you're giving additional training data. So we can skip over the whole problem of specifying the inputs and outputs and let the algorithm create a model that does the work for us. Now, when it comes to AVs, they will, in a deep sense, be powered by machine learning. You can't have an AV without it, using only traditional algorithms. This is not to say that autonomous vehicles are machine learners all the way down. They're often hybrid systems, but at least for example, vision or object recognition, they rely heavily on machine learning.

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Teaching a car to recognize an object as a stop sign, a double yellow line, a person using traditional programming techniques is not as reliable or accurate as machine learning based approaches.

Basl 23:16 What does this mean for the ethics of programming for accident scenarios? Well, the fact that machine learning is central to AV technologies has important implications for the ethics of AV design. As a first lesson, some people in the literature emphasize the importance of focusing on accident scenarios, because they think you need to program for every scenario. But that's not right. In this image, we can imagine that every point under the curve is a unique situation that an AV might find itself in. The curve itself tells us the probability that a particular event might occur. So as we see, accident scenarios that look like trolley cases are in a region that's very improbable. The orange area represents the most common scenarios an AV might be in and the shaded area represents a training set or a data lake that we might use to train the algorithm for object recognition or driving. keeping it simple, we'll just talk about driving behavior. Once we're in the machine learning paradigm, what we're doing is building a training set and allowing an algorithm to extrapolate to other scenarios. This is the foundation of the paradigm. The tools are affected because we don't try to cover every scenario. We can tinker with what that data lake looks like and what parts of the curve under the curve it draws from. But given that we need machine learning to develop AVs, we're already committed to not trying to explicitly program for every scenario under that curve. This doesn't show that trolley optimism is false, but it does undermine one motivation you might have for defending the view. Some might claim that it is negligent to ignore accident scenarios that are similar to trolley cases, even though they're improbable, but it isn't negligent. Machine learning is a useful technology precisely because we don't have to code for every case. It isn't throwing one's hands up or throwing in the towel on the problem. It's taking the only approach available for achieving some ends - safe, reliable autonomous driving. But the centrality of machine learning to AVs does more than undermine a motivation for trolley optimism. It grounds a case for trolley pessimism. When we think about how we want AVs to behave, we have to think about how to influence that behavior via training. And that means thinking holistically about the role that training plays in influencing driving behavior, and the various choice points in the process of training that involve asking ethical questions. Now, as it turns out, there are many questions that precede questions about whether we should program AVs to behave in particular ways in accident scenarios or that mirror particular vertex that are ethically relevant. To illustrate this idea, it's useful to distinguish a variety of questions we might ask about the relationship between accident scenarios, trolley cases and machine learning techniques. First, regime. How much of the training regime should be dedicated to dealing explicitly with accident scenarios? Should we have any accident scenarios in our training data? What proportion? Structure. Should accident scenarios in the training set be structured like trolley cases? And which trolley cases? There are lots of accident scenarios that don't look like trolley cases at all, from simple fender benders to more complex cases not involving trade offs between lives. And there are lots of variations of trolley cases. If we include some in our training regime, which ones and how should they be structured? Verdict. Should verdicts and trolley cases be used to assign rewards or punishments to responses and accident scenarios? Even if we think we should include some accident scenarios that look like trolley cases, should we use our vertex to give the thumbs up or the thumbs down or the reward or punishment? Or should we reward according to some other metric or behavior? Finally, behavior. Should AVs be programmed to behave in accident scenarios in ways that conform with the appropriate verdict in similarly structured trolley cases? Trolley optimists focus on behavior. Seeing AVs as technologies that rely on machine learning helps us to see that these other ethical, these other questions are relevant. Now, I said these questions precede the question that the optimist cares about answering. To see that, just notice that the answer to regime is none, that we should include no

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accident scenarios and the training regime. Then the answer to all these other questions is no, no and no. If we shouldn't dedicate any of our regime to accident scenarios at all, then the answer to structure verdict and behavior are settled.

Basl 27:27 Why might we think the answer to a question like regime is none? Well, let's briefly return to the example of dog training. Let's say you want your dog to do lots of things. But you're very concerned to ensure that they stay on command. Not just at home, not just at the dog park but everywhere you might use the command. To achieve that outcome, you'd have to provide lots of training data in lots of different environments, you'd have to give lots of examples and you'd have to spend a lot of your training regime, the time you have available for training the dog, to train that command, potentially at the cost of others. The same is potentially true in the context of ensuring that an AV behaves in a particular way in a particular kind of accident scenario. If you want to get a right in not just one trolley like accident scenario, but in all or most of them will have to make accident scenarios a significant focus of training efforts. But this could come at a cost. In the case of our dog, we spend all our time training stay, we might come home to a mess on the floor. Similarly, if we overtrain for accident scenarios that could come at a cost to everyday driving functionality. Now, we don't know what the trade offs might be. That's an empirical, technical question. It could turn out that training heavily for particular behaviors in accident scenarios succeeds at the cost of putting AVs into scenarios more frequently. So they get the answer right when they're in accidents in trolley-like accident scenarios, but they're in a lot more trolley-like accident scenarios. Or maybe it succeeds at the cost of an increase in more frequent minor accidents. It doesn't matter what the trade offs are. The point is that upon full consideration of the ethical question of regime and the possibility of such trade offs, we might decide that we should include no accident scenarios in our training regime. That all said, the case we're making doesn't hang on answering regime in any particular way or there being any actual trade offs. What matters for our purposes, and what the possibility of trade offs makes vivid is that regime is an ethical question, as are the other questions that precede behavior, and other similar ones like them. They invoke values and they require ethical principles to help resolve. And these ethical questions are the sorts of questions which it makes sense to apply the traditional tools of moral philosophy, at least in part. Now, one such tool is thought experiments and typically thought experiments are used to test cases for use as test cases for principles. Highlighting the limits of principles, helping us to refine, dispose of or adopt them. And trolley cases or thought experiments typically used to draw attention to for example, the relevance of total harms and the relevance of doing versus allowing harms and helping us to develop principles that are sensitive to the considerations. And they might be relevant here in our theorizing about how to answer ethically sensitive design questions like regime or others. We're not taking a stance on that. But what they don't do is entail an answer to behavior. The relevance of trolley cases to Annabel is that they can inform the way she thinks about the training program she ultimately ought to adopt. What the optimist wants us to do is to take a shortcut, going straight from judgments about trolley cases to what we should make a car do and accident scenarios that look like those trolley cases, ignoring any preceding design questions like regime and we recognize the temptation. If an AV in a trolley-like accident scenario veered away from one person to kill five other people, wouldn't that tell us that we made some deep mistake in answering our design questions or that we messed up the training regime? The answer is no. To see why, let's assume that after some serious theorizing, we come to take the lessons of trolley cases to be maximized, lives saved. Now we don't actually think this is the lesson from trolley cases but it does provide an illustrative example that will apply no matter what we take the lesson to be. Now on the surface, it may appear that this principle counsels us to abandon any machine learning regime that results in an AV hitting the five when it could have instead hit the one. On the

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contrary, though, this principle might actually entail that we are required to enact such a regime, so long as doing so minimizes deaths across all the driving contexts in which the AV operates. We can imagine that only these two design approaches are possible, dedicating half our training regime to accident scenarios or none of it. We can even stipulate that we know for sure that in design one, if a car was in a trolley like accident scenario, it would kill the five instead of the one. Design two on the other hand, would get it right in such a case. But design one might still yield a higher number of total lives saved depending on how much better drives and other scenarios compared to design two. The lesson we're after is that since there are ethical questions that precede and are in principle answerable independently of behavior, the fact that trolley cases are structurally similar to accident scenarios, does not by itself provide any reason to program AVs to behave in ways that mirror appropriate verdicts in those cases. If the answer to behavior turns out to be yes, it will be because of reasons we have for adopting the answers we do to regime, structure and verdict. And whatever reasons we have for answering these questions in a way that we do is not because of some structural similarity between trolley cases and AV behavior.

Basl 32:30 To close, let's consider a potential objection to our view, and then highlight some implications of our argument. Now, one type of objection seeks to undermine our argument by pointing out technological solutions that undermine our claim that we have to answer questions like regime before questions of behavior. For example, with infinite training data and infinite processing power, we don't have to worry about trade offs between different training regimes, we can trade for all possible scenarios and all possibilities without worry - there's no interaction between them. Now, obviously, this isn't a real solution, we don't have that much data or that much processing power. But the infinite training data solution does point the way to a kind of response. Infinite training data works as an objection to our view and our argument, because it isolates behavior and accident scenarios from other questions about AV design. It lets us essentially slip into the traditional programming paradigm, where we can think about what we should do with respect to a particular kind of behavior independently of others. So, are there any technical solutions that allow us to separate out accident scenarios from the rest of driving behavior? Well, we don't want to underestimate the cleverness of computer scientists or programmers or the people that will ultimately build these technologies in these vehicles. They have lots of tools in their toolkit, they can tinker with lots of variables but at the same time, we have to admit, we aren't particularly worried about this kind of objection. To see why, let's see how our objection survives a different kind of technological response - that of divide and conquer. Now, the divide and conquer approach attempts to break down difficult problems of training into parts. For example, highway driving presents different problems than city driving. And so you might want to train a model that can address the problems of highway driving separately from a model that addresses the problems of city driving. And so you train two separate models, and then you create a thing that tells it when to deploy the highway model versus the city model. And so there you divide and conquer the different problems. Well, why not apply this approach to AVs? You can have a model that is trained to manage typical driving scenarios versus a model that's designed explicitly to manage accident scenarios that look like trolley cases. So why doesn't this response work? Well, first, it's not really clear that this is a feasible approach. Unlike highway versus city driving, where it's easy to tell when to deploy which subroutine, you could use GPS data or things like that to say when to deploy the highway model versus the city model, we would most certainly have to train an algorithm or a model to decide whether to deploy the accident subroutine or the normal sub routine. But that model itself involves questions like regime. So we're just back to asking ethical questions that are prior to the questions of behavior and might have implications for it. But even if we could deploy the divide and conquer approach, this would just make it so that we could train a car to behave a certain way in accident scenarios. That

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doesn't make regime or other questions about training any less value laden. We still need to think about whether we should take the approach of divide and conquer. And that's an ethical question that we have to deploy the traditional tools of moral philosophy to. So let's just briefly mention some implications. First, there's the narrow implication that we can't read off how we should make AVs behave and accident scenarios directly from trolley cases. So our advice to Annabel shouldn't be to try to do so. More broadly, taking seriously the underlying features of the technologies that power AVs and other applications, transforms the ethical problems that we face. And we need to think really carefully about how to deploy ethical resources to these problems. We need to develop or figure out how the principles we develop in moral philosophy apply appropriately. Questions like regime or other questions about developing training for machine learning algorithms.